

PENZA, Russian Federation

WMO: 279620

Lat: 53.1228N

Lon: 45.0189E

Elev: 172

StdP: 99.27

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.1	(c) -23.7	(d) -30.0	(e) 0.2	(f) -26.9	(g) -26.5	(h) 0.3	(i) -23.4	(j) 10.9	(k) -3.4	(l) 10.0	(m) -4.1	(n) 2.3	(o) 220	(p) 0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.0	19.9	30.0	19.4	28.1	18.6	21.5	28.5	20.6	27.5	19.7	26.2	4.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.2	24.2	18.2	13.4	23.2	17.4	12.7	22.5	63.2	28.3	60.1	27.5	57.2	26.3	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.7	(b) 9.1	(c) 8.4	(e) -30.6	(f) 34.2	(g) 3.4	(h) 2.2	(i) -33.0	(j) 35.7	(k) -35.0	(l) 37.0	(m) -36.9	(n) 38.2	(o) -39.3	(p) 39.8		
(a) 9.7	(b) 9.1	(c) 8.4	(e) -30.2	(f) 23.3	(g) 3.6	(h) 1.2	(i) -32.8	(j) 24.1	(k) -34.9	(l) 24.8	(m) -37.0	(n) 25.5	(o) -39.6	(p) 26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-9.5	-8.9	-3.5	6.9	14.1	17.9	20.4	18.6	12.9	5.8	-1.3	-6.8		
	DBStd	12.00	7.04	6.96	5.21	5.34	4.86	4.23	3.57	4.06	4.45	4.68	5.50	6.80		
	HDD10.0	2721	603	530	418	123	17	2	0	0	20	146	340	522		
	HDD18.3	4849	862	763	677	343	147	59	20	47	171	389	590	781		
	CDD10.0	1122	0	0	0	30	144	238	322	266	107	15	0	0		
	CDD18.3	208	0	0	0	1	16	46	83	55	8	0	0	0		
	CDH23.3	2232	0	0	0	9	204	493	864	579	82	0	0	0		
	CDH26.7	754	0	0	0	0	46	160	320	213	15	0	0	0		
(14) Wind	WSAvg	3.9	4.1	4.4	4.3	3.9	4.1	3.6	3.1	3.3	3.5	4.0	4.2	4.2		
(15) Precipitation	PrecAvg	540	40	30	31	35	40	58	65	56	52	48	47	42		
	PrecMax	719	108	84	97	92	93	139	200	115	155	106	110	83		
	PrecMin	332	2	0	2	7	6	12	12	0	1	0	4	16		
	PrecStd	110	23	20	19	20	24	33	39	29	36	27	26	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	2.9	11.1	24.9	30.1	34.0	35.2	35.1	28.5	20.2	11.5	5.8		
		MCWB	2.6	1.8	7.1	12.9	17.2	19.4	19.9	20.4	16.4	12.7	9.7	4.9		
	2%	DB	1.2	1.8	6.8	21.8	28.0	30.5	32.1	31.5	25.6	17.1	9.4	2.9		
		MCWB	0.7	1.0	3.7	12.3	16.5	19.5	20.2	20.0	15.6	11.6	8.4	2.2		
	5%	DB	0.7	1.0	4.5	18.9	25.8	28.2	30.1	28.9	23.4	14.7	7.2	1.8		
		MCWB	0.2	0.4	2.4	10.9	16.0	18.7	19.9	19.0	15.2	11.1	6.2	1.3		
	10%	DB	-0.1	0.1	2.9	16.1	23.5	26.1	28.1	26.4	20.9	12.6	5.2	0.9		
		MCWB	-0.5	-0.4	1.4	9.4	14.8	18.1	19.5	18.1	14.1	9.8	4.5	0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.0	6.9	14.8	19.1	21.9	22.9	22.5	18.5	14.7	10.2	5.0		
		MCDB	2.7	2.4	10.7	20.8	26.8	29.2	29.6	29.9	23.4	17.0	11.0	5.3		
	2%	WB	0.9	1.1	4.2	13.2	17.7	20.7	21.7	21.0	17.1	12.9	8.3	2.2		
		MCDB	1.1	1.4	6.1	18.9	25.4	27.5	28.6	28.3	23.0	15.4	9.2	2.7		
	5%	WB	0.2	0.4	2.6	11.6	16.7	19.8	20.9	19.9	15.9	11.3	6.3	1.2		
		MCDB	0.5	0.8	4.0	17.8	24.3	26.2	27.7	27.1	21.3	14.1	7.0	1.5		
	10%	WB	-0.7	-0.4	1.5	10.1	15.5	18.7	20.1	19.0	14.8	10.1	4.5	0.5		
		MCDB	-0.3	0.2	2.5	15.6	21.8	24.6	26.5	25.2	19.8	12.2	5.1	0.8		
(35) Mean Daily Temperature Range	MDBR		6.0	7.3	7.8	10.3	12.0	11.4	11.5	11.7	10.3	7.5	5.1	5.5		
	5% DB	MCDBR	4.3	4.9	8.3	14.5	14.9	14.4	14.8	15.3	14.0	11.1	6.7	3.8		
		MCWBR	4.4	4.5	5.8	7.1	6.6	5.9	5.9	6.1	6.4	6.9	5.6	3.7		
	5% WB	MCDBR	4.2	4.9	7.0	13.2	13.3	12.5	12.8	13.4	12.3	9.7	5.9	3.9		
		MCWBR	4.3	4.8	5.2	7.2	6.5	5.9	5.9	6.0	6.5	6.6	5.3	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.326	0.359	0.426	0.393	0.387	0.404	0.427	0.377	0.354	0.326	0.296		
	taud		2.099	2.034	2.035	2.090	2.310	2.361	2.317	2.240	2.357	2.340	2.240	2.107		
	Ebn at Noon		651	743	806	799	856	865	840	793	794	726	617	564		
	Edh at Noon		72	106	131	143	120	116	119	122	96	79	64	60		
(44) All-Sky Solar Radiation	RadAvg		0.88	1.88	3.40	4.33	5.57	5.85	5.87	4.82	3.16	1.57	0.66	0.50		
	RadStd		0.17	0.30	0.44	0.42	0.48	0.54	0.50	0.42	0.43	0.23	0.14	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	N/A

Nomenclature: See separate page